

Answer Assignment 1_PERMUTATION & COMBINATION

$$Q1. {}^nP_r = \frac{n!}{(n-r)!}$$

$$(I) {}^8P_3 = {}^nP_r = \frac{n!}{(n-r)!} = \frac{8!}{(8-3)!} = \frac{8!}{5!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1} = 336$$

$$(II) {}^8P_8 = \frac{8!}{(8-8)!} = 8!$$

$$(III) {}^{30}P_2 = \frac{30!}{(30-2)!} = \frac{30!}{28!} = 30 \times 29 = 870$$

$$(IV) {}^{20}P_4 = \frac{20!}{(20-4)!} = \frac{20!}{16!} = 20 \times 19 \times 18 \times 17 = 116280$$

$$(V) {}^7P_4 = \frac{7!}{(7-4)!} = \frac{7!}{3!} = 7 \times 6 \times 5 \times 4 = 840$$

$$(VI) {}^{12}P_5 = \frac{12!}{(12-5)!} = \frac{12!}{7!} = 12 \times 11 \times 10 \times 9 \times 8 = 95040$$

$$(VII) {}^9P_9 = \frac{9!}{(9-9)!} = 9!$$

$$(VIII) {}^9P_5 = \frac{9!}{(9-5)!} = \frac{9!}{4!} = 9 \times 8 \times 7 \times 6 \times 5 = 15120$$

$$(IX) {}^{25}P_2 = \frac{25!}{(25-2)!} = \frac{25!}{23!} = 25 \times 24 = 600$$

$$(X) {}^{56}P_3 = \frac{56!}{(56-3)!} = \frac{56!}{53!} = 56 \times 55 \times 54 = 166320$$

$$Q.2 {}^nC_r = \frac{n!}{(r!)(n-r)!}$$

$$(I) {}^7C_3 = \frac{7!}{(3!)(7-3)!} = \frac{7!}{3!4!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$$

$$(II) {}^8C_4 = \frac{8!}{(4!)(8-4)!} = \frac{8!}{4!4!} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 70$$

$$(III) {}^{20}C_6 = \frac{20!}{(6!)(20-6)!} = \frac{20!}{6!14!} = \frac{20 \times 19 \times 18 \times 17 \times 16 \times 15}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 38760$$

$$(IV) {}^5C_1 = \frac{5!}{(1!)(5-1)!} = \frac{5!}{1!4!} = \frac{5}{1} = 5$$

$$(V) {}^6C_6 = \frac{6!}{(6!)(6-6)!} = \frac{6!}{6!0!} = 1$$

$$(VI) {}^7C_2 = \frac{7!}{(2!)(7-2)!} = \frac{7!}{2!5!} = \frac{7 \times 6}{2 \times 1} = 21$$

$$(VII) {}^{10}C_4 = \frac{10!}{(4!)(10-4)!} = \frac{10!}{4!6!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

$$(VIII) {}^{50}C_{50} = \frac{50!}{(50!)(50-50)!} = \frac{50!}{50!0!} = 1$$

$$(IX) {}^{60}C_{58} = \frac{60!}{(58!)(60-58)!} = \frac{60!}{58!2!} = \frac{60 \times 59}{2 \times 1} = 1770$$

$$(X) {}^{25}C_{21} = \frac{25!}{(21!)(25-21)!} = \frac{25!}{21!4!} = \frac{25 \times 24 \times 23 \times 22}{4 \times 3 \times 2 \times 1} = 12650$$

4. Sol. Required number of 5 digit number formed if no digit is repeated = ${}^5P_5 = \frac{5!}{(5-5)!} = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

(b) Sol. Required number of 5 digit number formed if repetition of digit is allowed = $5^5 = 5 \times 5 \times 5 \times 5 \times 5 = 3125$

5. Sol. Required number of 4 digit number formed if no digit is repeated = ${}^4P_4 = \frac{4!}{(4-4)!} = 4! = 4 \times 3 \times 2 \times 1 = 24$

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6. Sol. Required number of 5 digit number formed = $4! \times 4 = 4 \times 3 \times 2 \times 1 \times 4 = 96$

7. Sol. Required number of 5 letters formed if repetition is not allowed = ${}^5P_5 = \frac{5!}{(5-5)!} = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

8. Sol. (a) $H \underline{O} T \underline{E} L = 2! \times 4! = 2 \times 24 = 48$

(b) Sol. Total number of words formed by using all the letters of the given words = $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Number of words each having vowels together = $H \underline{O} T \underline{E} L = 2! \times 4! = 2 \times 24 = 48$

Number of words each having vowels never together = $120 - 48 = 72$

9. The group has 6 letters of which T occurs twice.

Had all the digits been different, the answer would have been ${}^6P_6 = 6!$

Since T occurs twice, so the number of ways arranging the letter would be

$$\frac{6!}{2!} = \frac{6 \times 5 \times 4 \times 3 \times 2!}{2!} = 360$$

10. The group has 10 letters of which I occur twice. Had all the letters been different, the answer would have been ${}^{10}P_{10} = 10!$

However, since I occurs twice, so the number of ways arranging the letter would be

$$\frac{10!}{2!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 1814400$$

11. We have total seven letters therefore, $n = 7$ and words to be formed are also of seven letters, therefore, $r = 7$

$${}^7P_7 = \frac{7!}{(7-7)!} = 7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$$

12. Sol. Required number of letters formed if repetition is not allowed = $\frac{9!}{3!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 60480$

13. (a) Sol. $F \underline{I} \underline{E} L D = 2! \times 4! = 2 \times 24 = 48$

(b) Sol. Total number of words formed by using all the letters of the given words = $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Number of words each having vowels together = $F \underline{I} \underline{E} L D = 2! \times 4! = 2 \times 24 = 48$

Number of words each having vowels never together = $120 - 48 = 72$

(c) Sol. $F _ _ _ = 1! \times 4! = 1 \times 24 = 24$

Ans: 3

(d) Sol. $_ _ _ D = 4! \times 1! = 1 \times 24 = 24$

(e) Sol. $_ _ \underline{E} _ _ = 1! \times 4! = 1 \times 24 = 24$

(f) Sol. $F _ _ _ D = 1! \times 3! \times 1! = 6$

14. Sol. Required number of 5 letters formed if repetition is not allowed = ${}^8P_5 = \frac{8!}{(8-5)!} = 8 \times 7 \times 6 \times 5 \times 4 = 6720$

15. Sol. Required number of 4 letters formed if repetition is not allowed = ${}^7P_4 = \frac{7!}{(7-4)!} = 7 \times 6 \times 5 \times 4 = 840$

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16. Sol. Required number of letters formed if repetition is not allowed = ${}^7P_2 =$

$$\frac{7!}{(7-2)!} = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 840$$

17. Sol. Required number of letters formed if repetition is not allowed = ${}^8P_8 =$

$$\frac{8!}{(8-8)!} = 8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$$

18. Sol. Required number of ways = ${}^7P_7 =$

$$\frac{7!}{(7-7)!} = 7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$$

19. Sol. Required number of ways = $(7-1)! = 6! = 720$

20. Sol. Required number of ways = ${}^9C_6 =$

$$\frac{9!}{3!6!} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

(b) Sol. Required number of ways = ${}^{12}C_6 =$

$$\frac{12!}{6!6!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 924$$

21. Sol. Required number of ways = ${}^7C_3 \times$

$${}^6C_3 = \frac{7!}{3!4!} \times \frac{6!}{3!3!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 35 \times 20 = 700$$

(b) Consisting of two men and four women? A:315

Sol. Required number of ways = ${}^7C_2 \times {}^6C_4 =$

$$\frac{7!}{2!5!} \times \frac{6!}{4!2!} = \frac{7 \times 6}{2 \times 1} \times \frac{6 \times 5}{2 \times 1} = 21 \times 15 = 315$$

22. Sol. Total no of handshakes = ${}^{18}C_2 =$

$$\frac{18!}{2!16!} = \frac{18 \times 17}{2 \times 1} = 153$$

23. Sol. Total no of handshakes = ${}^{35}C_2 =$

$$\frac{35!}{2!33!} = \frac{35 \times 34}{2 \times 1} = 595$$

24. Sol. Required number of ways = $7! \times 6! = 5040 \times 720 = 3628800$

25. Sol. Required number of ways = $5! \times 4! = 120 \times 24 = 2880$

26. Sol. Required number of ways = $3! \times 7! = 6 \times 5040 = 30240$

27. Sol. 4 Gents around a circular table = $(4-1)! = 3! = 6$ and

4 ladies in 4 ways = $4! = 24$

Required number of ways = $6 \times 24 = 144$

28. Sol. Required number of ways = $5! \times 8! = 120 \times 40320 = 4838400$

29. Sol. Required number of ways = $8! \times 4! = 40320 \times 24 = 967680$

30. A (i) 462 (ii) 330

31. (i) A :462 (ii) 462

32. Sol: Number of words each having vowels together = W A T E R = $2! \times 4! = 2 \times 24 = 48$

33. Sol: Number of words each having vowels together = P H R A S E = $2! \times 5! = 2 \times 120 = 240$

34. Number of words each having vowels together = N A T I Q N A L = $\frac{4!}{2!} \times \frac{5!}{2!} = 12 \times 60 = 720$

35. Sol: Number of words each having vowels together = P H A N T O M = $2! \times 6! = 2 \times 720 = 1440$

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36. Sol. Total number of words formed by using all the letters of the given words = $7!$
 $= 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$

Sol: Number of words each having vowels together = $P H \underline{A} N T \underline{O} M = 2! \times 6! = 2 \times 720 = 1440$

Number of words each having vowels never together = $5040 - 1440 = 3600$

37. Sol: Number of words each having vowels together = $\underline{O} R G \underline{A} N \underline{I} Z \underline{E} D = 4! \times 6! = 24 \times 720 = 17280$

38. Sol: Number of words each having vowels together = $M \underline{A} G N \underline{I} T \underline{U} D \underline{E} = 4! \times 6! = 24 \times 720 = 17280$

39. 2

40. (i) Required number of ways = $2! \times 4! = 2 \times 24 = 72$

(ii) After selection of Amit, we are left with 3 persons (excluding Ishan) out of which 2 are to be selected.

$\therefore$ Total no. of required ways = $1 \times {}^3C_2 = 1 \times 3 = 3$.

41.

(1 st digit)	(2 st digit)	(3 st digit)	(4 st digit)	(5 st digit)
Any one of 3, 6 or 8				

3P_1				
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3P_1 ways and remaining 4 digits $\frac{4!}{2!}$ (as 0 occurs twice) ways

$${}^3P_1 \times \frac{4!}{2!} = \frac{3 \times 4 \times 3 \times 2!}{2!} = 36 \text{ ways}$$

42.

(1 st digit)	(2 st digit)	(3 st digit)	(4 st digit)	(5 st digit)
Only 7, 8 or 9				
3P_1				

3P_1 ways and remaining 4 digits $\frac{4!}{3!}$ ways

$${}^3P_1 \times \frac{4!}{3!} = \frac{3 \times 2!}{2!} \times \frac{4 \times 3 \times 2 \times 1}{1} = 72 \text{ ways}$$

43.

(1 st digit)	(2 st digit)	(3 st digit)
Only 5 or 6		
2P_1		

2P_1 ways and remaining 6 digits 6P_2 ways

${}^2P_1 \times {}^6P_2 =$ Therefore total numbers are = $2 \times 30 = 60$ ways

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Number of 4 digits less than 3000 is

(1 st digit)	(2 st digit)	(3 st digit)	(4 st digit)
Only 1 or 2			
2P_1			

2P_1 ways and remaining 6 digits 6P_3 ways

${}^2P_1 \times {}^6P_3$ = Therefore total numbers are =
 $2 \times 120 = 240$ ways

Required numbers are = $60 + 240 = 300$

44. (a) [Total no. of committees (from 9 men & 5 women)] - [No. of committee without women (only men)]

$${}^{14}C_4 - {}^9C_4 = \frac{14 \times 13 \times 12 \times 11}{4 \times 3 \times 2 \times 1} - \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 1001 - 126 = 875$$

(b) Sol: Total no. of ways = ${}^9C_4 \times {}^5C_4 =$
 $\frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} \times \frac{5}{1} = 126 \times 5 = 630$

45: Total no. of ways
 $= {}^7C_6 + ({}^4C_1 \times {}^7C_5) + ({}^4C_2 \times {}^7C_4) + ({}^4C_3 \times {}^7C_3)$
 $= 7 + 84 + 210 + 140 = 441$

46. Answer: 6 gentlemen, 4 Lecturer; committee of 5.

At least one Lecturer to be included; the combinations are:

(1L, 4P), or (2L, 3P), or (3L, 2P), or (4L, 1P)

$${}^4C_1 \times {}^6C_4 + {}^4C_2 \times {}^6C_3 + {}^4C_3 \times {}^6C_2 + {}^4C_4 \times {}^6C_1 = 6 + 120 + 60 + 6 = 246.$$

2nd Method:

[Total no. of committees (from 6 Professors & 4 Lecturers)] - [No. of committee without Lecturers (only Professors)]

$${}^{10}C_5 - {}^6C_5 = \frac{10 \times 9 \times 8 \times 7 \times 6}{1 \times 2 \times 3 \times 4 \times 5} - \frac{6}{1} = 252 - 6 = 246$$

47. Sol: Total no. of ways
 $= {}^7C_6 + ({}^7C_5 \times {}^5C_1) + ({}^7C_4 \times {}^5C_2) = 462$

48. A) 6 B) 9

49. a) $5! \times 2! = 240$

b) $3! \times 4! = 144$

50. $2! \times 3! \times 5! = 1440.$